Review Problems 4

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In preparing for midterm exams review your lecture notes, assigned course readings, problem sets and prepare answers to the questions on the review sheet(s). You may bring to the exam a single 8.5×11 inch sheet of paper with notes on it. No calculation aides are allowed in the exam (e.g., calculators, phones, laptops etc.). You may work in pencil or pen, however I advise the use of pencil. Please be sure to bring sufficient blue books with you to the exam if possible. Scratch paper will be provided.

[1] [Exploitation, Exploration and Bandits] In each of $t = 1, 2$ rounds an agent needs to pull either Arm 0 or Arm 1. Let $D_t = 0$ if the agent pulls Arm 0 in round t and $D_t = 1$ if she pulls Arm 1 instead. Let $Y_t \in \{0, 1\}$ denote the observed reward/payoff in round t. Ideally, the agent would like to maximize the total expected payoff across the two rounds (i.e., the expectation of the sum $Y_1 + Y_2$). The agent has the following priors over the properties of the two arms. Let θ_d denote the unknown success rate for arms $d = 0, 1$.

1. Arm 0: The agent knows, with certainty, that $\theta_0 \in \Theta_0 = \left\{\frac{1}{4}, \frac{3}{4}\right\}$. *A priori* she believes that $\Pr\left(\theta_0 = \frac{1}{4}\right) = \frac{1}{4}$ and $\Pr\left(\theta_0 = \frac{3}{4}\right) = \frac{3}{4}$. Arm 0 is either "good" ($\theta_0 = \frac{3}{4}$) or "bad" ($\theta_0 = \frac{1}{4}$).
2. Arm 1: The agent knows, with certainty, that $\theta_1 \in \Theta_1 = \{\kappa, 1 - \kappa\}$ for some $0 < \kappa < 1$. *A priori* she believes that $\Pr(\theta_1 = 0) = \frac{1}{2}$ and $\Pr(\theta_1 = 1) = \frac{1}{2}$. Arm 1 is either "nearly perfect" ($\theta_1 = 1 - \kappa$) or "almost broken" ($\theta_1 = \kappa$).

The two priors are independent. Let $\mathcal{I}_t$ denote the begining-of-period t information set of the agent. At $t = 1$ the agent has no information such that $\mathcal{I}_1 = \emptyset$ (she just has her prior beliefs, π). In the second round ($t = 2$) the information set includes the choices and outcomes from round 1 (i.e, $\mathcal{I}_2 = (D_1, Y_1)'$).

[a] A priori what is the expected per round payoff of each arm? That is compute

$$\Pr(Y_1 = 1 | D_1 = 0) = \mathbb{E}[Y_1 | D_1 = 0] = \boxed{?} \quad \Pr(Y_1 = 1 | D_1 = 1) = \mathbb{E}[Y_1 | D_1 = 1] = \boxed{?}$$

Which arm would a myopic agent choose in round 1? Explain **[1 to 2 sentences].**

[b] Assume the agent pulls arm 0 in round $t = 1$. Calculate the agent's posterior beliefs in the event that this pull is a failure ($Y_1 = 0$) as well as when it is a success ($Y_1 = 1$):

$$\Pr\left(\theta_0 = \tfrac{3}{4} \middle| \mathcal{I}_2 = (0,0)'\right) = \boxed{?} \quad \Pr\left(\theta_0 = \tfrac{3}{4} \middle| \mathcal{I}_2 = (0,1)'\right) = \boxed{?}$$
$$\Pr\left(\theta_0 = \tfrac{1}{4} \middle| \mathcal{I}_2 = (0,0)'\right) = \boxed{?} \quad \Pr\left(\theta_0 = \tfrac{1}{4} \middle| \mathcal{I}_2 = (0,1)'\right) = \boxed{?}$$

(Hint: *You only need to explicitly compute two of the four beliefs above since "probabilities sum to one".*)

[c] Assume the agent instead pulls arm 1 in round $t = 1$. Calculate the agent's posterior beliefs in the event that this pull is a failure ($Y_1 = 0$) as well as when it is a success ($Y_1 = 1$):

$$\Pr\left(\theta_1 = 1 - \kappa \middle| \mathcal{I}_2 = (1,0)'\right) = \boxed{?} \quad \Pr\left(\theta_1 = 1 - \kappa \middle| \mathcal{I}_2 = (1,1)\right) = \boxed{?}$$
$$\Pr\left(\theta_1 = \kappa \middle| \mathcal{I}_2 = (1,0)'\right) = \boxed{?} \quad \Pr\left(\theta_1 = \kappa \middle| \mathcal{I}_2 = (1,1)'\right) = \boxed{?}$$

[d] Observe that there are four possible histories defining the agent's beginning-of-round $t = 2$ information set (i.e., $\mathcal{I}_2 = (D_1, Y_1)' \in \left\{(0,0)', (0,1)', (1,0)', (1,1)'\right\}$). These histories reflect round $t = 1$ choices and

payoff outcomes. Your research assistant uses you answers from parts [b] and [c] above to compute the expected payoff from pulling arm 0 in round $t = 2$ given each of these four possible histories,

$$\begin{aligned}
\mathbb{E}\left[Y_2 \middle| D_1 = 0, Y_1 = 0, D_2 = 0\right] &= \frac{1}{2} \\
\mathbb{E}\left[Y_2 \middle| D_1 = 0, Y_1 = 1, D_2 = 0\right] &= \frac{7}{10} \\
\mathbb{E}\left[Y_2 \middle| D_1 = 1, Y_1 = 0, D_2 = 0\right] &= \frac{5}{8} \\
\mathbb{E}\left[Y_2 \middle| D_1 = 1, Y_1 = 1, D_2 = 0\right] &= \frac{5}{8}
\end{aligned}$$

as well as the the corresponding payoffs associated with pulling arm 1 in round $t = 2$,

$$\begin{aligned}
\mathbb{E}\left[Y_2 \middle| D_1 = 0, Y_1 = 0, D_2 = 1\right] &= \frac{1}{2} \\
\mathbb{E}\left[Y_2 \middle| D_1 = 0, Y_1 = 1, D_2 = 1\right] &= \frac{1}{2} \\
\mathbb{E}\left[Y_2 \middle| D_1 = 1, Y_1 = 0, D_2 = 1\right] &= 2\kappa - 2\kappa^2 \\
\mathbb{E}\left[Y_2 \middle| D_1 = 1, Y_1 = 1, D_2 = 1\right] &= 1 - 2\kappa + 2\kappa^2
\end{aligned}$$

Briefly explain how your research assistant computed these expected payoffs. **[2 to 3 sentences]** Which of these expectations are invariant to the payoff outcome of round 1? Explain. **[2 to 3 sentences]**

[e] Use part [d] to determine the agent's expected payoff maximizing round $t = 2$ arm choice, denoted by D_2^* under each of the four possible histories? That is, complete the table:

Arm\Outcome	$Y_1 = 0$	$Y_1 = 1$
$D_1 = 0$	$D_2^* = \boxed{?}$	$D_2^* = \boxed{?}$
$D_1 = 1$	$D_2^* = \boxed{?}$	$D_2^* = \boxed{?}$

Is optimal behavior in round $t = 2$ forward-looking? Explain. **[1 to 2 sentences]**

[f] Let Y_2^* denote an agent's payoff when they choose their round $t = 2$ arm optimally, conditional on their round $t = 1$ arm choice and payoff outcome. Explain why – when the agent chooses arm 0 in round $t = 1$– her expected round $t = 2$ payoff if she behaves optimally is

$$\begin{aligned}
\mathbb{E}\left[Y_2^* \middle| D_1 = 0\right] = \sum_{y_1=0,1} \sum_{d_2^*=0,1} &\{\mathbb{E}\left[Y_2 \middle| D_1 = 0, Y_1 = y_1, D_2^* = d_2^*\right] \\
&\times \Pr\left(D_2^* = d_2^* \middle| D_1 = 0, Y_1 = y_1\right) \times \Pr\left(Y_1 = y_1 \middle| D_1 = 0\right)\}
\end{aligned}$$

[2 to 4 sentences] (<u>Hint</u>: *What "law" is used?*) Use your answers in parts [a], [b], [c], [d] and [e] above to numerically evaluate the expression above. Similarly compute the expected optimal round $t = 2$ payoff conditional on pulling arm 1 in round $t = 1$ instead:

$$\begin{aligned}
\mathbb{E}\left[Y_2^* \middle| D_1 = 1\right] = \sum_{y_1=0,1} \sum_{d_2^*=0,1} &\{\mathbb{E}\left[Y_2 \middle| D_1 = 1, Y_1 = y_1, D_2^* = d_2^*\right] \\
&\times \Pr\left(D_2^* = d_2^* \middle| D_1 = 1, Y_1 = y_1\right) \times \Pr\left(Y_1 = y_1 \middle| D_1 = 1\right)\}
\end{aligned}$$

[g] Based on your answer in part [f], which arm would a forward-looking agent choose in round $t = 1$?

What would her total payoff be? Why does the choice of such an agent differ (or not differ) from the behavior of the myopic agent? In what sense is foregoing exploitation in order to explore a feature of optimal choice? **[4 to 6 sentence]**.

[h] Let $\lambda = \frac{\Pr(Y_1=1|D_1=0)}{\Pr(Y_1=1|D_1=0)+\Pr(Y_1=1|D_1=1)}$ and $U \sim \text{Uniform}[0,1]$. A "bandit" chooses arm 0 in round $t = 1$ if $U \leq \lambda$ and arm 1 otherwise. In round $t = 2$ the bandit becomes greedy and chooses the arm with the highest posterior expected payoff. Explain why this bandit will, in expectation, earn more that a myopic agent, but less than a forward looking one. **[3-6 sentences]**